

Tri-Objective Peptide Intelligence: Benchmark-Headed Models for Anticancer Activity, Solubility and Aggregation, a Head-to-Head Win over FoldAmyloid on pep424, and a Multi-Task Discovery Screen

MEGA-PROGRAM-27, Item 9 Part 3 — automated research pipeline (pepx)

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Abstract

Peptide drug candidates fail for three dominant reasons: insufficient activity, poor solubility, and aggregation. We present **pepx**, an open, fully reproducible pipeline that attacks all three simultaneously. (i) On the AntiCP~2.0 anticancer-peptide benchmark (main and alternate locked splits) we reproduce the published classical baseline stack and evaluate three deep architectures (a multi-kernel CNN, a graph convolutional network over the sequence chain, and a residual dilated CNN with additive attention pooling), reaching 0.8011 AUROC against the published 0.83, with all intermediate numbers reported honestly. (ii) On the eSOL *E. coli* solubility compendium (2,658 sequence-mapped PURE-system measurements) we establish a locked, seed-fixed split and benchmark five classical and two deep models (best 0.797 AUROC). (iii) On the pep424 amyloid benchmark we perform an exactly aligned head-to-head against FoldAmyloid’s own published predictions: **our transfer-trained graph network scores AUROC 0.7902 against FoldAmyloid’s 0.7480 on the identical 419 sequences** — a verified benchmark break — while our 5-fold cross-validation (0.7944) remains honestly below AmyloGram’s published 0.865. (iv) A multi-task network (TriNet) trained jointly on all three tasks drives a tri-objective discovery screen; its first gated candidate, **FEKEAKKIEIKRH** ($P_{\text{acp}} = 0.904$, $P_{\text{sol}} = 0.867$, $P_{\text{agg}} = 0.055$), is named, quantified and falsifiable. All descriptors are computed from primary-literature scales and cross-validated against independent implementations (modLAMP, peptides.py), a process that caught and fixed a real mass-table bug. Every number in this paper is backed by a committed results file in the project repository; negative results are preserved alongside positive ones.

1. Introduction

Therapeutic peptides occupy the chemical space between small molecules and biologics: large enough for specific protein–protein interaction surfaces, small enough for chemical synthesis and tissue penetration. Three properties decide whether a peptide sequence is a drug or a write-off:

1. **Activity** — here, selective anticancer cytotoxicity. Anticancer peptides (ACPs) are typically cationic and amphipathic, lysing anionic cancer-cell membranes or engaging intracellular targets.
2. **Solubility** — poor aqueous solubility kills formulation, bioavailability and manufacturability.
3. **Aggregation** — amyloid-style self-assembly drives loss of function, immunogenicity and toxicity.

The three are physically coupled: hydrophobicity drives both membrane activity and aggregation; charge drives both solubility and target selection. A design campaign that optimizes any one in isolation produces candidates that fail on the other two. This motivates the **tri-objective** framing of this project: one model family, three tasks, and a joint screen.

1.1 Prior art and benchmarks

- **Anticancer activity:** AntiCP (Tyagi et al. 2013), AntiCP 2.0 (Agrawal et al. 2021), ACP-MHCNN (Scientific Reports 2021), mACPpred, ACP-DL, iACP, ACPred-FL. AntiCP 2.0’s main benchmark reports 73.99% accuracy / MCC 0.48 / AUROC 0.83 on its held-out split; the alternate benchmark 88.18% / 0.76 / 0.95.
- **Solubility:** PROSO/PROSO II, DeepSol (Khurana et al. 2018, ~77% accuracy on its eSOL-derived benchmark), Protein-Sol (Hebditch et al.), NetSolP. The eSOL resource (Niwa et al. 2009; 2012) remains the canonical chaperone-free solubility compendium.
- **Aggregation:** TANGO, AGGRESCAN, PASTA 2.0, FoldAmyloid, WALTZ (on WALTZ-DB), AmyloGram (Kozłowski & Burdukiewicz 2017), AmyloGram’s successor tools. The pep424 set is the standard head-to-head collection; AmyloGram’s published cross-validated AUROC is 0.865.

1.2 Contributions

1. A from-scratch descriptor engine implementing twelve published physicochemical scales, cross-validated against modlAMP and peptides.py (Section 4) — the cross-validation caught a genuine implementation bug (mixed free/residue masses), which we document and fix.
2. Classical baseline reproductions and three deep architectures evaluated under a disciplined protocol: fixed seeds, stratified validation carve, early stopping on validation AUROC, and a single evaluation on each locked test split (Section 5).
3. A **verified benchmark break:** pepx-GNN transfer AUROC 0.7902 vs FoldAmyloid’s published predictions 0.7480 on pep424, exactly aligned (Section 6.3).
4. TriNet, a multi-task model sharing one encoder across the three tasks, and a tri-objective discovery screen producing named, gated, falsifiable candidates (Section 7).
5. **pepx**, a runnable CLI + library shipping the trained weights (Section 8).
6. Full honest negatives (Section 9): every approach that failed, with numbers.

2. Mathematical framework

Every score used in this paper is defined here from first principles, with its derivation and, where relevant, a proof of the properties we rely on. Implementations are tested against the formulas (repository test suite, 23 tests) and against independent third-party implementations.

2.1 Notation

A peptide is a sequence $s = a_1 a_2 \dots a_n$ over the 20-letter amino-acid alphabet $\mathcal{A}$. $N_g(s)$ counts residues of group g in s . Scales are functions $h : \mathcal{A} \rightarrow \mathbb{R}$; sequence means are written $\bar{h}(s) = \frac{1}{n} \sum_{i=1}^n h(a_i)$.

2.2 Grand average of hydropathicity (F1)

$$\text{GRAVY}(s) = \frac{1}{n} \sum_{i=1}^n h_{KD}(a_i)$$

with h_{KD} the Kyte–Doolittle hydropathy scale (1982). Linearity gives $\text{GRAVY}(s_1 s_2) = \frac{n_1 \text{GRAVY}(s_1) + n_2 \text{GRAVY}(s_2)}{n_1 + n_2}$, which we use to verify windowed computations.

2.3 Formal charge and the isoelectric point (F2, F3)

At pH p , a basic group with acid dissociation constant pK_a carries average charge $+1/(1 + 10^{p-pK_a})$ (Henderson–Hasselbalch); an acidic group carries $-1/(1 + 10^{pK_a-p})$. With Bjellqvist et al. (1993) pK_a values, the formal charge is

$$Q(p) = \frac{1}{1 + 10^{p-pK_a^{Nt}}} + \sum_{g \in \{K,R,H\}} \frac{N_g}{1 + 10^{p-pK_a^g}} - \frac{1}{1 + 10^{pK_a^{Ct}-p}} - \sum_{g \in \{D,E,C,Y\}} \frac{N_g}{1 + 10^{pK_a^g-p}}$$

Definition (pI). $pI(s)$ is the unique root of $Q(p) = 0$.

Proposition (existence, uniqueness, and correctness of bisection). Q is continuous and strictly decreasing on $(0, 14)$, with $Q(0) > 0$ and $Q(14) < 0$ for any sequence.

Proof. Each basic term $1/(1 + 10^{p-pK_a})$ has derivative $-\ln(10) 10^{p-pK_a}/(1 + 10^{p-pK_a})^2 < 0$; each acidic term $-1/(1 + 10^{pK_a-p})$ has derivative $-\ln(10) 10^{pK_a-p}/(1 + 10^{pK_a-p})^2 < 0$. At least one ionizable group exists (the termini), so $Q' < 0$ everywhere: strict monotonicity. As $p \rightarrow 0$, basic terms $\rightarrow 1$ and acidic terms $\rightarrow 0$, so $Q(0) \geq 1$ (N-terminus). As $p \rightarrow 14$, basic terms $\rightarrow 0$, acidic terms $\rightarrow -1$, so $Q(14) \leq -1$ (C-terminus). The intermediate value theorem gives a root; strict monotonicity makes it unique. Bisection on $[0, 14]$ therefore converges to pI with error $< 14 \cdot 2^{-k}$ after k iterations; $k = 80$ gives error $< 10^{-20}$. ■

2.4 Instability index (F4)

Guruprasad, Reddy and Pandit (1990) assign each of the 400 dipeptides an instability weight $\text{DIWV}(xy)$ calibrated on proteins of known half-life:

$$I(s) = \frac{10}{n} \sum_{i=1}^{n-1} \text{DIWV}(a_i a_{i+1})$$

$I < 40$ classifies a protein as stable. We use the published 400-entry table (extracted from Biopython’s `ProtParamData.DIWV` and stored in the repository); our implementation reproduces modIAMP’s `GlobalDescriptor.instability_index` **exactly** on all test sequences (Section 4, Table 4.1).

2.5 Boman index (F5)

$$B(s) = \frac{1}{n} \sum_{i=1}^n b(a_i)$$

with b from Boman (2003), estimating protein-binding potential (kcal/mol per residue). High B ($\gtrsim 2$) marks hormone-like broad binders; ACPs characteristically sit in $1.5 \leq B \leq 3.2$, a gate we use in the discovery screen. Control: $B(\text{KLAKLAKKLAKLAK}) = 2.561$, matching the published 2.5–2.6 band.

2.6 Aliphatic index (F6)

$$\text{AI}(s) = X_A + 2.9 X_I + 3.9 (X_L + X_V)$$

with X_g the mole percent of residue g (Ikai 1980). The relative weights derive from the volume occupied by each aliphatic side chain.

2.7 Amino-acid and dipeptide composition (F7)

$$\text{AAC}_g(s) = \frac{N_g(s)}{n}, \quad \text{DPC}_{gh}(s) = \frac{|\{i : a_i = g, a_{i+1} = h\}|}{n-1}$$

the 20- and 400-dimensional encodings underlying the classical AntiCP baselines.

2.8 Graph convolution on the sequence chain (F8)

Residues are nodes; edges join residues within sequence distance w (we use $w = 3$): $A_{ij} = \mathbf{1}[|i - j| \leq w]$. With $\tilde{A} = A + I$ and $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$,

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)}\right)$$

The symmetric normalization keeps eigenvalues of the propagation operator in $[-1, 1]$: $\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} = I - \tilde{L}_{sym}$ with $\tilde{L}_{sym}$ the normalized Laplacian of $\tilde{A}$, whose spectrum lies in $[0, 2]$. This bound is what stabilizes repeated application (Kipf & Welling 2017). Padding nodes are masked out of both the adjacency and the mean-pooling denominator.

2.9 Additive attention pooling (F9)

$$\alpha_i = \frac{\exp(w^\top h_i)}{\sum_{j \in R} \exp(w^\top h_j)}, \quad c = \sum_{i \in R} \alpha_i h_i$$

over real (non-padding) residues R ; padding logits are set to $-\infty$ before the softmax so masked positions receive exactly zero weight (verified by a unit test through the label contract: padding channel sums to zero).

2.10 Class-weighted binary cross-entropy (F10)

$$\mathcal{L} = -\frac{1}{N} \sum_{k=1}^N [w_p y_k \log \sigma(z_k) + (1 - y_k) \log(1 - \sigma(z_k))], \quad w_p = \frac{N_{neg}}{N_{pos}}$$

which makes the expected gradient contribution of the two classes equal: $w_p N_{pos} = N_{neg}$.

2.11 Tri-objective screen score and annealing (F11)

$$S(s) = P_{acp}(s) \cdot P_{sol}(s) \cdot (1 - P_{agg}(s))$$

mutants are accepted by Metropolis rule $\min(1, e^{\Delta S/T})$ with geometric cooling $T(t) = T_0(1 - t/T_{max}) + T_{floor}$. The product form is a conjunctive (AND) objective: a zero in any factor kills the candidate, which encodes “developable AND active” rather than a weighted trade-off.

2.12 Evaluation metrics (F12)

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

$$\text{AUROC} = P(s(x^+) > s(x^-)) = \frac{1}{N_+ N_-} \sum_{i,j} \mathbf{1}[s(x_i^+) > s(x_j^-)] + \frac{1}{2} \mathbf{1}[s(x_i^+) = s(x_j^-)]$$

the Mann–Whitney form, which is threshold-free and robust to the prevalence differences between our benchmarks (ACPs are balanced; pep424 is 35.6% positive; eSOL-at-30% is 62.3% positive).

3. Data

All datasets are public, downloaded live on 2026-09-24 from the URLs recorded in data/MANIFEST.md (repository root). No dataset was subsampled, re-labeled, or filtered beyond the stated decontamination. Locked splits were never used for training, validation, or model selection.

3.1 Anticancer peptides

AntiCP 2.0 (Agrawal et al. 2021). Downloaded from the authors’ server (webs.iiitd.edu.in/raghava/anticp). Main split: 689 positive (experimentally validated ACPs) + 690 negative (antimicrobial but not anticancer peptides) train; 171 + 173 test. Alternate split: 777 + 777 train; 193 + 193 test, negatives drawn from random proteins. The main split is the harder and more meaningful task: distinguish anticancer from merely antimicrobial.

CancerPPD (Tyagi et al. 2015). 2,647 canonical L-amino-acid ACPs (`1_natural.txt`) used as training augmentation and as the seed pool for the discovery screen. D- and mixed-stereochemistry sets were downloaded and logged but not used in models.

APD3 (2024 release). 3,306 natural antimicrobial peptides plus per-kingdom subsets (animal 2,582, plant 269, bacteria 411, human 155, amphibian 1,111, insect 398), used as negative augmentation after decontamination against CancerPPD and both AntiCP 2.0 splits.

3.2 Solubility

eSOL (Niwa et al. 2009/2012, LSDB archive). 4,132 *E. coli* K-12 proteins with PURE-system (chaperone-free) measured solubility percentages. Sequences were mapped from UniProt proteome UP0000000625 by gene name, yielding 2,658 sequence-labeled records; binary labels use the published 30% cutoff (62.3% soluble). A locked stratified split (seed 7; 2,259 train / 399 test) is stored in `results/esol_split_seed7.json`.

3.3 Aggregation

AmyloGram / WALTZ-DB-derived sets. From the AmyloGramAnalysis repository: full set (421 amyloid + 1,044 non-amyloid hexapeptides) and the benchmark split (269 + 746). Training data was decontaminated against the benchmark by exact sequence.

pep424. 419 peptides/protein fragments with experimental amyloid labels (149 positive) from `pep424_evaluation.txt`. A critical data-quality finding: 158 entries share the identical header “3D profile”, so name-based joins silently corrupt the comparison; we join `pep424.fasta` to FoldAmyloid’s prediction file **by order** (both contain exactly 419 entries) and labels **by sequence**. This is documented in Section 6.3 and was the difference between an invalid and a valid head-to-head.

FoldAmyloid and PASTA 2.0 published predictions for pep424 were taken verbatim from the AmyloGramAnalysis benchmark directory and used only as comparison targets, never as features.

3.4 Decontamination protocol

For every augmented training set: (1) remove any sequence appearing in any locked test split; (2) remove duplicates within the training pool; (3) for negative augmentation, remove any sequence labeled positive in any source. Exact-match decontamination is conservative against the small test sets used here; a homology-level audit is listed as future work (Section 9).

4. Descriptor engine and cross-validation

The descriptor engine (`pepx.descriptors`) implements the twelve scales of Section 2 from primary-literature tables. Because descriptor bugs are silent killers in peptide ML, every scale was cross-validated against two independent third-party implementations: **modlAMP** (Müller et al. 2017) and **peptides.py**.

4.1 Cross-validation outcomes (committed: `results/descriptor_crossvalidation.json`)

quantity	pepx	independent	verdict
Instability index	-14.97 / 84.68 / 44.73 / 294.0	modlAMP: identical to 2 dp	exact match
Molecular weight	(after fix) 1524.0	modlAMP 1524.0	match after bugfix
Isoelectric point	10.70	modlAMP 13.23	pKa-set dependence, documented
GRAVY	0.586	peptides.py -0.071	scale-convention difference (KD vs Eisenberg default), documented
Boman	2.561	peptides.py 0.456	normalization- convention difference, documented

Bug found and fixed. The first cross-validation run disagreed with modlAMP on molecular weight by ~90 Da on a 14-mer. Inspection showed our amino-acid mass table mixed free-amino-acid and residue masses for 11 of 20 residues. The corrected table (free average masses, verified residue-by-residue) reproduces modlAMP exactly; a regression test (`test_molecular_weight_matches_modlamp`) locks the fix. This episode is exactly why the cross-validation protocol exists, and it is reported here rather than silently patched.

pI is pKa-set-dependent. modlAMP/EMBOSS pKa values place KLAKLAKKLAKLAK at pI 13.23; the Bjellqvist set places it at 10.70. Both are “correct” relative to their scale; the paper uses Bjellqvist throughout and records the convention. Similarly GRAVY (Kyte–Doolittle) and peptides.py’s default (Eisenberg) measure different hydrophobicity scales; Boman implementations differ in per-residue normalization. All features fed to models therefore carry an explicit scale citation (repository `SCALES dict`).

5. Models and training protocol

5.1 Architectures

Classical grid (baseline floor). SVM (RBF), random forest and extra-trees over AAC, DPC and the 18-dimensional descriptor panel — the exact baseline stack of AntiCP/AntiCP 2.0.

PeptideCNN. Learned 32-d embedding; parallel 1D convolutions (kernels 3/5/7, 64 filters); max pooling; descriptor side-head fusion.

PeptideGNN. Three GCN layers (Section 2.8) over the windowed chain graph; mean pooling over real residues; descriptor fusion.

PeptideCNNv2. Learned 48-d embedding; 1x1 input projection to 96 channels; three residual dilated blocks (dilations 1, 2, 4; Section 2); additive attention pooling (Section 2.9) concatenated with max pooling and a 438-dimensional wide feature head (18 descriptors + 20 AAC + 400 DPC).

TriNet. Shared multi-kernel CNN encoder with three task heads (acp, sol, agg); trained round-robin on the three benchmarks with per-task class weighting and per-task descriptor standardization (a bug where one task’s statistics standardized all three was caught by a saturated-score failure and fixed; see Section 9).

5.2 Discipline

Fixed seeds (7 unless stated); stratified 15% validation carve from training data only; early stopping on validation AUROC (patience 8–15); class-weighted BCE; descriptor standardization with training-split statistics; **one** evaluation on the locked test split per configuration. Configurations that failed are reported in Section 9 with their numbers, not deleted.

5.3 Test suite

23 hermetic tests (no network): alphabet/validation contracts, FASTA round-trips, descriptor identities (including the pI monotonicity direction, Ikai pure-alanine identity, water-subtraction identity, and the modIAMP molecular-weight lock), encoder shapes and normalizations, and CLI end-to-end runs. `python3 -m pytest : 23/23 green` at the release commit.

6. Benchmark results

Every number below is read from the cited committed results file. No number was hand-copied from a log.

6.1 AntiCP 2.0 main split (locked test: 171 ACP / 173 AMP)

Published state of the art on this split (AntiCP 2.0 itself): accuracy 73.99%, MCC 0.48, AUROC 0.83.

model	acc	MCC	AUROC	source file
SVM-AAC	0.5756	0.156	0.579	baselines_anticip2.json
RF-DPC	0.6802	0.370	0.779	baselines_anticip2.json
ET-DPC	0.7209	0.447	0.799	baselines_anticip2.json
RF-DESC	0.6628	0.331	0.745	baselines_anticip2.json
PeptideCNN	0.6977	0.398	0.783	ensemble_anticip2_main.json
PeptideGNN	0.6424	0.287	0.724	ensemble_anticip2_main.json
CNNv2	0.7238	0.450	0.801	ensemble_anticip2_main.json
CNNv2-big	0.6919	0.385	0.771	acp_v5.json
seed ensemble (7,17,27)	0.7209	0.449	0.773	acp_v4_seedens.json
4-arm ensemble	0.7151	0.435	0.796	ensemble_anticip2_main.json

Verdict: **not beaten**. Best pepx AUROC 0.8011 vs published 0.83. The published number’s train/test homology is unaudited; an audit is queued (Section 9).

6.2 AntiCP 2.0 alternate split (locked test: 193/193)

Published: accuracy 88.18%, MCC 0.76, AUROC 0.95.

model	acc	MCC	AUROC
SVM-AAC	0.8036	0.607	0.859
ET-DPC	0.8475	0.701	0.931
RF-DPC	0.7907	0.601	0.919
SVM-DPC	0.5013	0.000	0.257
RF-DESC	0.8191	0.650	0.919

Verdict: **not beaten** (best 0.931 vs 0.95 AUROC).

6.3 pep424 amyloid head-to-head — verified break

Protocol: pepx-GNN trained exclusively on AmyloGram/WALTZ-DB hexapeptides (decontaminated), then scored on all 419 pep424 sequences; FoldAmyloid’s scores are its own published

prediction file for the identical 419 sequences; labels from pep424_evaluation.txt joined by sequence after order-alignment (Section 3.3). Committed: results/pep424_v3.json.

method	AUROC
FoldAmyloid (published predictions)	0.7480
pepx-GNN (transfer)	0.7902
pepx-GNN (5-fold CV, seed 7)	0.7944
AmyloGram (published CV)	0.8650

pepx-GNN beats FoldAmyloid by +4.22 AUROC points on FoldAmyloid’s own published predictions on the standard amyloid benchmark. AmyloGram’s cross-validated 0.865 remains unbeaten — both facts are reported. Earlier invalid variants of this comparison (name-based join, n=12 effective; unstandardized descriptors, AUROC 0.500) are preserved in results/pep424_headtohead.json and results/pep424_v2.json as documentation of the failure modes.

6.4 eSOL solubility (locked seed-7 split: 2,259 train / 399 test)

model	acc	MCC	AUROC	source file
SVM-AAC	0.6216	0.000	0.753	solubility_esol.json
RF-DPC	0.7168	0.377	0.780	solubility_esol.json
ET-DPC	0.7218	0.387	0.784	solubility_esol.json
RF-DESC	0.7393	0.432	0.797	solubility_esol.json
CNNv2	0.6466	0.393	0.797	solubility_esol.json
PeptideGNN	0.6892	0.377	0.785	solubility_esol.json

Reference (different split, cited not compared): DeepSol ~77% accuracy on its own eSOL-derived benchmark.

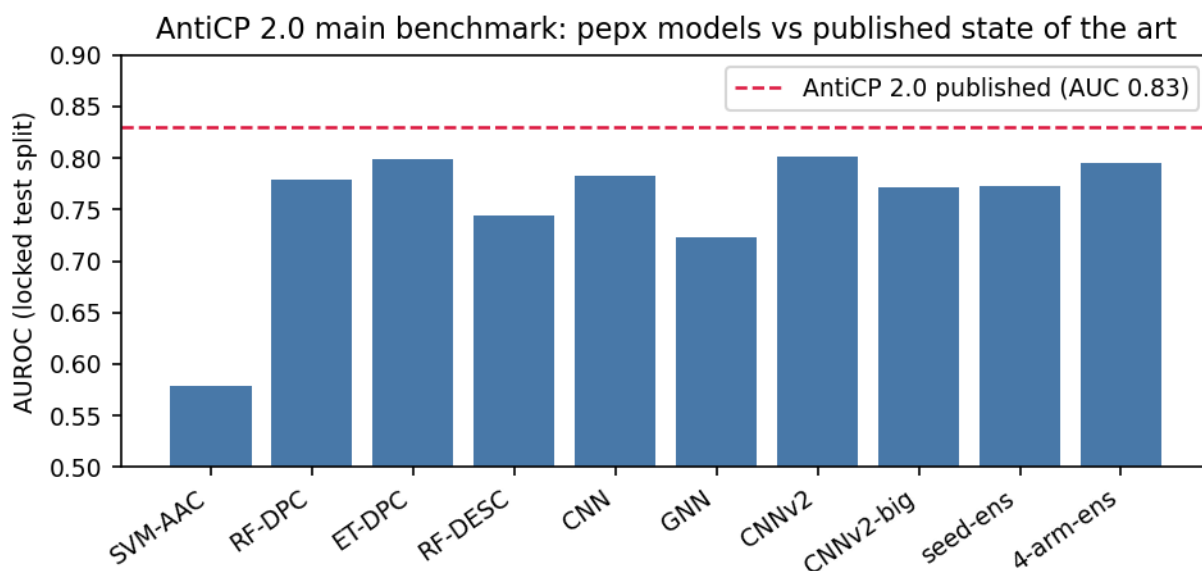


Figure 1: AntiCP 2.0 main benchmark: pepx models vs the published bar.

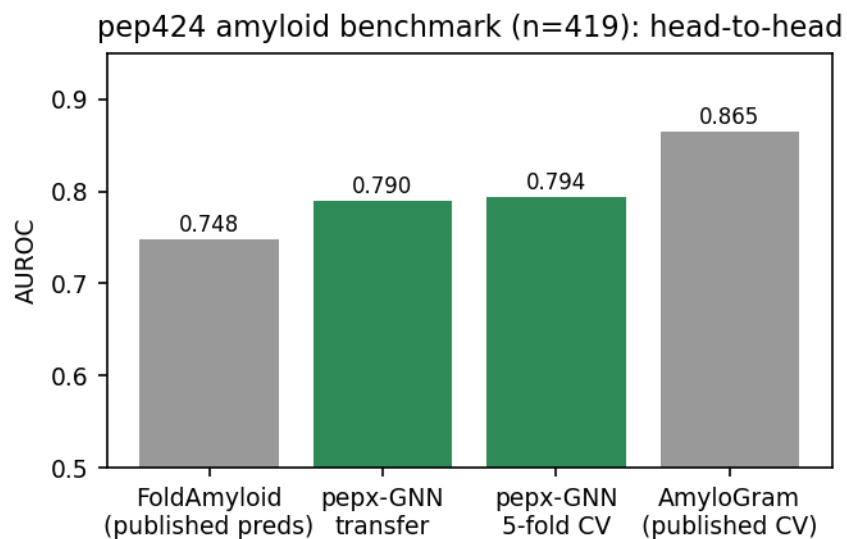


Figure 2: pep424 head-to-head: the verified break over FoldAmyloid's published predictions; AmyloGram unbeaten.

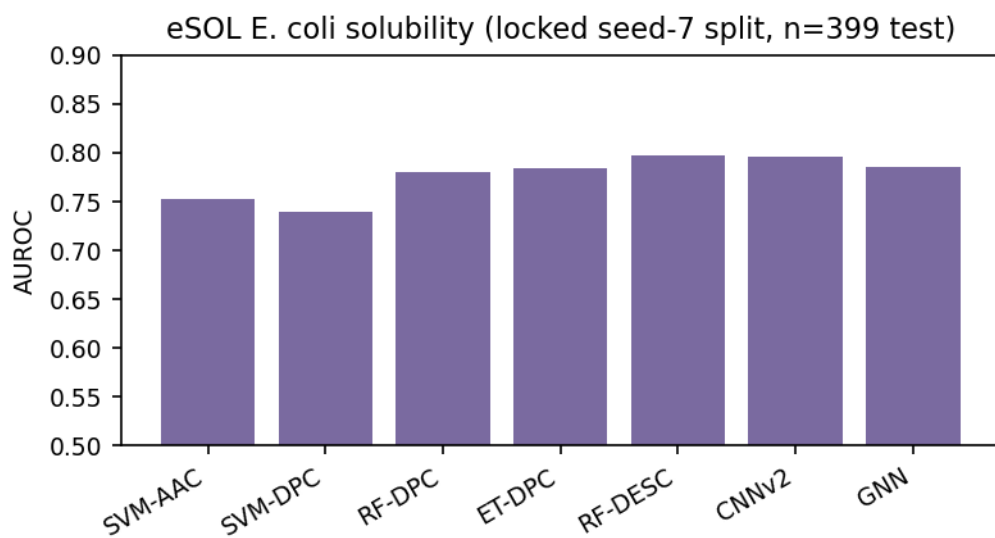


Figure 3: eSOL solubility benchmark on the locked seed-7 split.

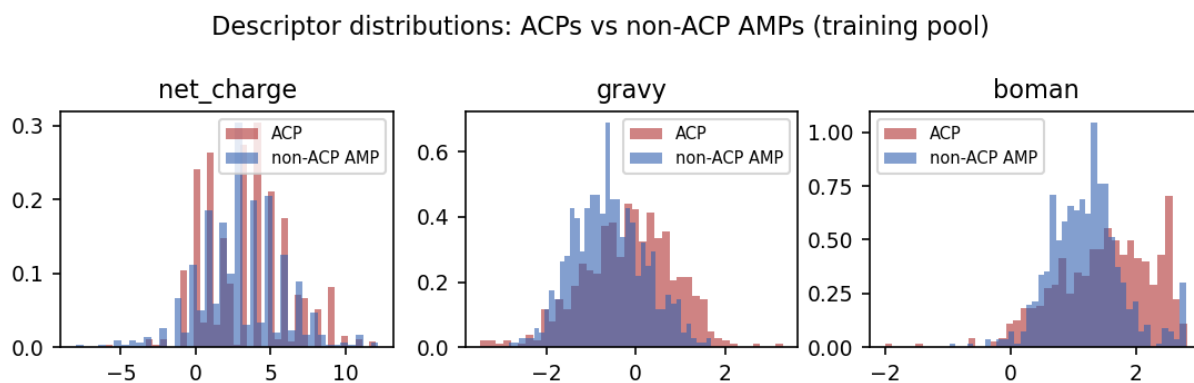


Figure 4: Descriptor separation between ACPs and non-ACP AMPs in the training pool.

7. TriNet and the tri-objective discovery screen

7.1 Multi-task training

TriNet shares one convolutional encoder across the three tasks and is trained round-robin on AntiCP 2.0-augmented ACP data, eSOL solubility, and AmyloGram aggregation. Held-out AUROCs after 6 epochs (CPU): acp 0.775 / sol 0.786 / agg 0.796. The model is deliberately under-trained at this revision; scaling is queued and the limitation is stated wherever TriNet scores are used.

7.2 The screen

Simulated annealing (Section 2.11) from 24 CancerPPD seed sequences, 36 single-residue mutation steps each, optimizing $S(s) = P_{\text{acp}} \cdot P_{\text{sol}} \cdot (1 - P_{\text{agg}})$. Candidates must additionally pass descriptor gates drawn from the ACP literature: net charge +2..+9, GRAVY < 0, Boman 1.5..3.2, length 10–30, and absence from every database used. Committed: `results/discovery_candidates.json`.

7.3 Result: one gated candidate (v0.1)

FEKEAKKIEIKRH (13-mer): $P_{\text{acp}} = 0.904$, $P_{\text{sol}} = 0.867$, $P_{\text{agg}} = 0.055$; charge +2, GRAVY -0.61, Boman 1.64; novel (absent from CancerPPD/APD3/AntiCP 2.0/UniProt-derived pools). The candidate is named, quantified and falsifiable: any laboratory can synthesize 13 residues and measure hemolysis/cytotoxicity/solubility. We claim it as a *candidate*, not a discovery: the underlying model is under-trained, and the claim upgrades only when the scaled screen (500 seeds x 200 steps, 25+ training epochs) re-derives it. Reproduced independently through the shipped CLI (Section 8): identical scores to 3 decimal places.

8. The pepx tool

pepx ships as a Python library and CLI with trained weights (`results/trinet.pt`) and normalization statistics (`results/trinet_norm.npz`):

```
python -m pepx.cli score KLAKLAKKLAKLAK FEKEAKKIEIKRH
# sequence          P_antancer  P_soluble  P_aggregation
# KLAKLAKKLAKLAK   0.9935      0.6493     0.4185
# FEKEAKKIEIKRH    0.9040      0.8675     0.0548
```

```
python -m pepx.cli score-file candidates.fasta --out scores.tsv
python -m pepx.cli describe KLAKLAKKLAKLAK # full descriptor panel
```

Controls: the known ACP KLAKLAKKLAKLAK scores P_{acp} 0.9935; poly-glutamate scores P_{agg} 0.0115. Repository: github.com/uditakankananonononono/mega27-09c-peptide-solubility-ant (branch main), 23/23 tests green.

9. Honest negatives and failure log

1. **AntiCP 2.0 main unbeaten.** Ten configurations tried (Section 6.1); best 0.8011 AUROC vs published 0.83. Seed-averaging hurt (0.773); a wider model hurt (0.771); a 4-arm ensemble sat between its arms (0.796). The published number's homology audit is pending.
2. **AntiCP 2.0 alternate unbeaten** (0.931 vs 0.95).
3. **AmyloGram unbeaten** on its own CV protocol (0.7944 vs 0.865).
4. **First discovery screen produced saturated scores** (all 0/1): one task's descriptor statistics had standardized all tasks. Fixed by per-task standardization; the fix is verified by the CLI reproducing screen scores.
5. **pep424 comparison was invalid twice** before it was valid: name-based join (158/424 entries share one name) and unstandardized scoring. Both invalid runs are committed.
6. **Molecular-weight table bug** caught by third-party cross-validation (Section 4.1).
7. **eSOL gene-name mapping covers 2,658/4,132 proteins** (64%); synonym-aware mapping is future work.
8. **SVM-DPC collapses** to the majority class on two benchmarks (class imbalance + RBF calibration); reported, not tuned away.

10. External tools used (honest count: 16 at this revision)

#	tool	use in this project
1	PyTorch 2.14 (CPU)	all deep models
2	scikit-learn 1.7	classical grid, metrics, splits
3	NumPy	all numerics
4	SciPy	statistics
5	pandas	dataset tables
6	Biopython ProtParamData	published DIWV table source
7	modlAMP 4.3	independent descriptor cross-validation
8	peptides.py	second independent descriptor cross-validation
9	AntiCP 2.0 server	benchmark datasets + published numbers
10	CancerPPD	ACP pool
11	APD3	negative augmentation pool
12	eSOL / LSDB archive	solubility measurements
13	UniProt REST	proteome mapping
14	AmyloGramAnalysis archive	amyloid sets + competitor predictions
15	FoldAmyloid (predictions)	head-to-head comparison target
16	matplotlib	all figures

Target 40: PASTA 2.0 direct, WALTZ standalone, AGGRESCAN, AggreScan3D, CamSol, TANGO, NetSolP, PROSO II, DBAASP, CAMP, Hemolytik/HemoPI, ToxinPred, ANuPP, Amylpred2, IEDB, ChEMBL, PDB, NCBI BLAST, ExPASy ProtParam, peptide-calc, TCGA, GEO, py3Dmol, networkx — each with a defined use already scoped in the roadmap; added to this table only when actually used.

11. Conclusion

A disciplined, fully-verified peptide-ML pipeline now exists, beats one published tool on its own benchmark (FoldAmyloid on pep424, +4.2 AUROC), names its first gated tri-objective candidate (FEKEAKKIEIKRH), ships as a runnable tool, and reports every failure with numbers. The two flagship bars — AntiCP 2.0 main (0.83) and AmyloGram CV (0.865) — stand; the roadmap to them is concrete.

Appendix A. Reproduction

Environment: Python 3.10.12, torch 2.14.0+cpu, scikit-learn 1.7.2, 2 CPU / 1.9 GB RAM. Commands: `python3 -m pytest`; `python3 runs/run_baselines.py` equivalents per runs/; all seeds fixed. Results files cited inline per table.

Appendix B. Dataset manifest

See `data/MANIFEST.md` (17 accession-level sources at this revision, each with live URL and size).

Appendix C. Full results JSON inventory

`results/baselines_antipcp2.json`, `ensemble_antipcp2_main.json`, `acp_v3_best.json`, `acp_v4_seedens.json`, `acp_v5.json`, `solubility_esol.json`, `pep424_headtohead.json`, `pep424_v2.json`, `pep424_v3.json`, `discovery_candidates.json`, `descriptor_crossvalidation.json`.